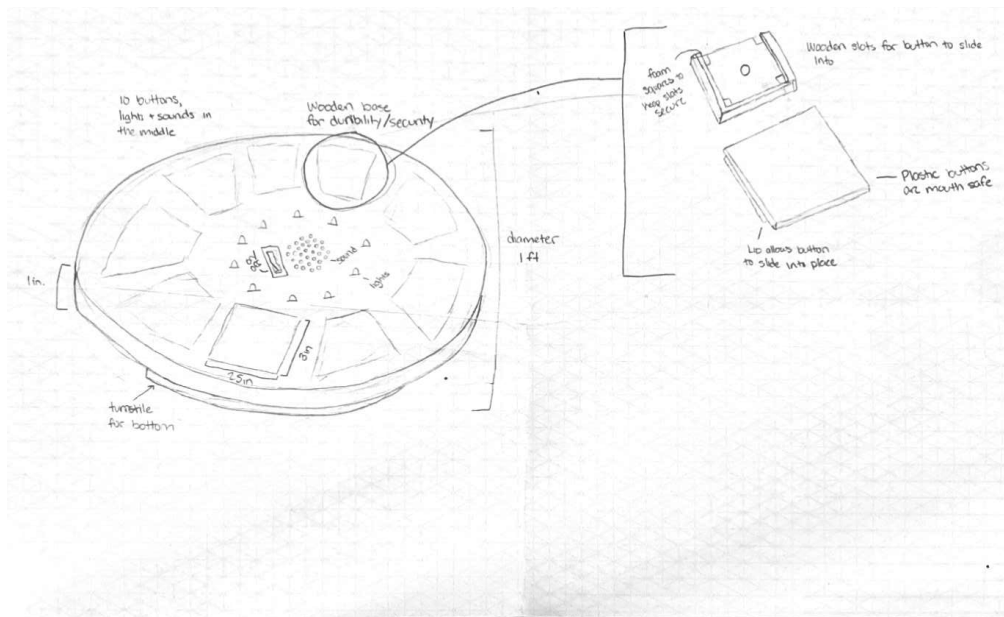


Element D

Morphological Chart

Functions:	Concept 1	Concept 2	Concept 3	Concept 4
Wiring	Raspberry pi	Arduino software	Orange pi	
Must have stimulating reward	Sounds go off when match is made	Light goes off when match is made	Different textures cover the device as stimulation when device is off	Device vibrates when match is made
Mouth safe	Pieces are large enough that they won't fit in the mouth	Device and buttons are plastic	Silicon coating on device and pieces	
Adjustable (buttons can be specialized for students)	Buttons slide in and out, and there are different sets of matches	Extra set of tiles to exchange for increased academic difficulty depending on student	Buttons are removable, extra buttons in case of loss	Buttons stay the same, velcro used to specialize
Can be turned on and off by a button or switch	On/off switch away from the buttons	On/off switch on base	On/off switch on side	On/off switch on bottom of the device
Mobile	Round, flat shape	Hand held device; under 1' tall	Cube shaped	Standing tower-like device under 2 ft
Buttons can still be pressed by students with motor disabilities	Buttons have a large surface area	Buttons are raised off device	Buttons are large and raise off platform	Buttons are spaced far apart
Pieces are secured to the device	Buttons are secured in wooden slots	Buttons rest on device	Buttons are clicked into place and can be removed with purpose	Buttons can be velcroed to device
Number of tiles	10 tiles on device at a time	16 tiles on device at a time	8 tiles on one specific side of the device	12 tiles on device at a time

Design #1:



This design is a round, spinning sensory toy that utilizes sound and lights as stimulating rewards. The student must spin the device in order to see all the matches. When the two buttons that are the correct match are pushed consecutively the device rewards the student with stimulating lights and sounds. The device spins through use of a turnstile and the buttons are secured on three sides with wooden slats.

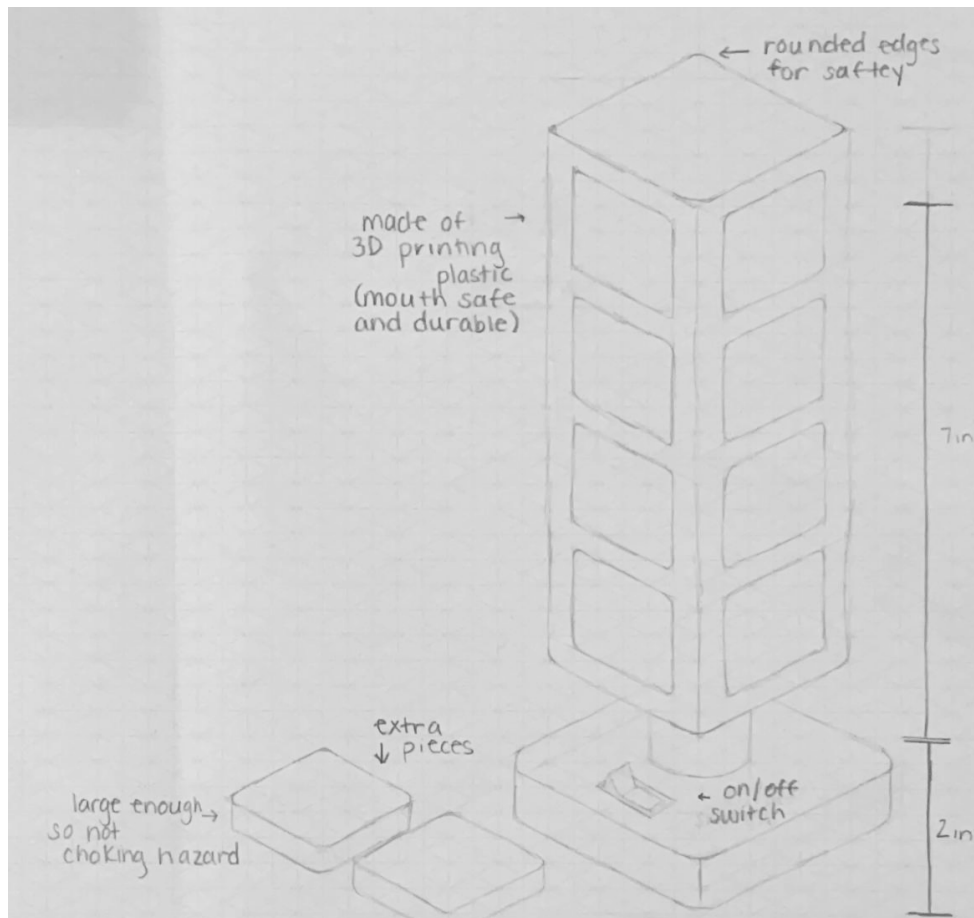
Advantages:

- Spins, so it is difficult for the kids to memorize the pattern
- Buttons are secured
- Buttons are large enough that kids with motor disabilities can press them
- Utilizes sound and lights as a sensory response
- Students are used to the round shape of the device and won't throw

Disadvantages:

- Students with motor disabilities may have difficulty spinning the device
- May be difficult to put in and take out buttons

Design #2:



This design was inspired from a conversation had with the teachers at Whitmarsh. They were discussing the viability of a vertical device to test the students vertical mobility skills. The design has buttons on all four sides and rotates around the center axel. This allows “masking” of buttons which touches another part of education in that class. When the matches are made a light at the top goes off as the reward.

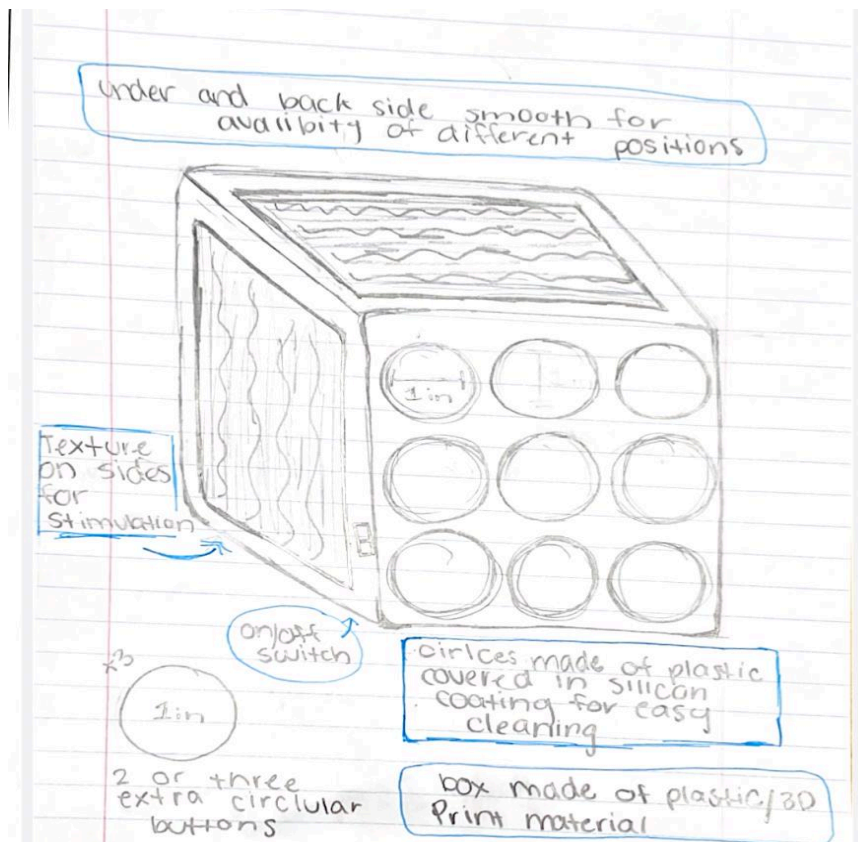
Advantages:

- Hand-held
- Buttons are large enough to be pushed by students with motor disabilities
- 16 buttons, with other to switch out for extra combinations
- “Masking items” because of its multiple sides

Disadvantages:

- May present issues with buttons staying in place because of vertical build
- Students with motor disabilities may face struggles with vertically stacked buttons

Design #3:



This product is a cube shaped cause and effect design. It utilizes a silicone coating on all removable pieces (10 total) to make it mouth safe and water resistant. Sound and light would both be used as a stimulating reward. Textures and other sensory materials would be located on the side of the design to make the cube multipurpose.

Advantages:

- Hand held/lightweight
- Buttons are large and easy to push
- Easy to clean
- Encourages education using light and sound

Disadvantages:

- Durability/balance
- Limited amount of buttons
- Buttons may be too close to one another
- Able to be thrown

Decision Matrix

		Design #1		Design #2		Design #3	
Qualities of Design	Weight	Score	Weighted Score	Score	Weighted Score	Score	Weighted Score
Have a Stimulating Reward	13%	5.0	0.7	5.0	0.7	5.0	0.7
Educational Value	10%	5.0	0.5	5.0	0.5	5.0	0.5
Adjustable	16%	4.0	0.6	5.0	0.8	4.0	0.6
Safety	13%	5.0	0.7	5.0	0.7	5.0	0.7
Mouth Safe	13%	5.0	0.7	5.0	0.7	5.0	0.7
Mobile	5%	3.5	0.2	5.0	0.3	3.5	0.2
On/Off Switch	5%	5.0	0.3	5.0	0.3	5.0	0.3
Buttons Can Be Pressed	7%	5.0	0.4	5.0	0.4	5.0	0.4
Pieces are secured	7%	5.0	0.4	2.0	0.1	3.5	0.2
Number of Tiles	2%	5.0	0.1	5.0	0.1	5.0	0.1
Test Motor Skills	2%	5.0	0.1	5.0	0.1	3.0	0.1
Durability	5%	5.0	0.3	4.0	0.2	4.0	0.2
Aesthetics	2%	4.0	0.1	3.0	0.1	5.0	0.1
Total:	100%		4.7		4.7		4.6

Rationale for weighting of categories:

To begin weighing the categories the group made sure that the high requirements of the design were weighted at least 10% as each design MUST address them and with diligence. The last 35% was divided between the low requirements of the design. The highest rated column is the adjustability of the device and it is that way because if the device is not able to be altered to the needs of the teachers and students it becomes minimally useful. The lowest rate columns are aesthetics and number of tiles, this is because the way the device looks is numerously less important than its functionality. The number of tiles is also low in importance because the devices need just 4 minimum buttons and all designs easily achieved it.

Rationale of choice:

Even though our decision matrix did not lead us to just one design, we chose design 2. This is because of the input from the teachers at Whitmarsh Elementary. When presenting our ideas to the teachers, they showed more interest in a vertical design, as they said it would incorporate another idea they have been working on in class. This idea is parallel play, meaning the students are getting used to playing with or even just playing near other students. With this in mind the group will be making a few design alterations to make sure the toy can accommodate multiple sets of tiles running simultaneously and that its structure will be equipped to handle multiple sets of hands and pressure.

MARCH						
SUNDAY	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY
					1 ← Testing	2
3 Status report #10	4	5	6 Element H due	7	8	9
10 Status report #11	11	12 Element I due	13 Testing at → Whitemarsh	14	15	16
17 Status report #12	18	19	20 Element J due	21	22	23
24 Status report #13	25	26	27	28	29 Element K due Element L due	30

Procedure:

Katie will focus on the coding and electronics aspect of the design as she has more experience with it from a previous engineering project. Maya and Mallory will meanwhile be designing the actual base of the design. This includes using CAD and laser printing and 3D printing pieces of the base design. Once it is time to assimilate both parts of the design the group will work together during testing and adjustments.

